

Towards the Unification of Haplotype Phasing and Assembly

Developing a versatile, undirected graphical
model for the single-individual haplotyping
problem

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Contents

1	Introduction	3
1.1	Background	3
1.1.1	State of DNA Sequencing	3
1.1.2	Importance of Haplotyping	3
1.1.3	Haplotype Assembly	4
1.1.4	Haplotype Phasing	4
1.2	Motivation	5
1.3	Problem Definition	5
2	Methods	6
2.1	Problem Inputs and Preliminaries	6
2.2	Building the Graphical Model	6
2.2.1	Fragment Graph	6
2.2.2	Quotient Graph	7
2.2.3	Factor Graph	8
2.3	Chordal Contraction	8
2.4	Inference	9
3	Results	11
3.1	Proof of Uniqueness	11
4	Discussion	13
5	Acknowledgements	14

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Abstract

Genome assembly methods are currently able to produce accurate genome assemblies, but due to the difficulty of concurrently assembling genomes and reconstructing haplotypes, these assemblies lack haplotype information. A haplotype is defined as a sequence of alleles that reside on the same chromosome. Accurately resolving haplotypes has many valuable and practical applications in population genetics, epigenetics, and inherited diseases. Determining haplotype phasings in humans requires assigning genetic alleles, or single nucleotide polymorphisms to the proper copy of the genome. There are two techniques used for computational haplotyping - haplotype assembly and haplotype phasing. Haplotype assembly is a combinatorial method which considers short-read DNA sequence information, and more recently, long-read sequences, to assemble haplotypes. Haplotype phasing is a probabilistic method which considers the individual's genotype and a database of compatible reference haplotypes, and typically leverages Markov chains to sample probable haplotypes. We propose a model inspired by the haplotype assembly and haplotype phasing problems that combines the complementary strengths of a combinatorial and probabilistic approach. This model is an undirected graphical model that computes the haplotype pair with the maximum likelihood from DNA sequence reads and reference haplotype information. Towards developing this model, we prove that upon joining two unique separate sets of haplotype phasings on a common variant site, the resulting combined phasings are also unique, a property that is necessary for efficient inference computation on the model.

1 Introduction

1.1 Background

1.1.1 State of DNA Sequencing

Current methods for DNA sequencing produce highly accurate sequence reads. Both next-generation short-read sequencing (NGS) done by Illumina and third-generation long-read sequencing (TGS) performed by PacBio and Oxford Nanopore continue to improve in read accuracy [12]. These sequencing techniques often leave haplotype phasing information unresolved, however [6]. Humans, as diploid organisms, carry two copies of each chromosome, which each have their own haplotype. These haplotypes are the set of single nucleotide polymorphisms (SNPs), or locations on the genomes which vary from human to human, that are inherited together on the same chromosome [17].

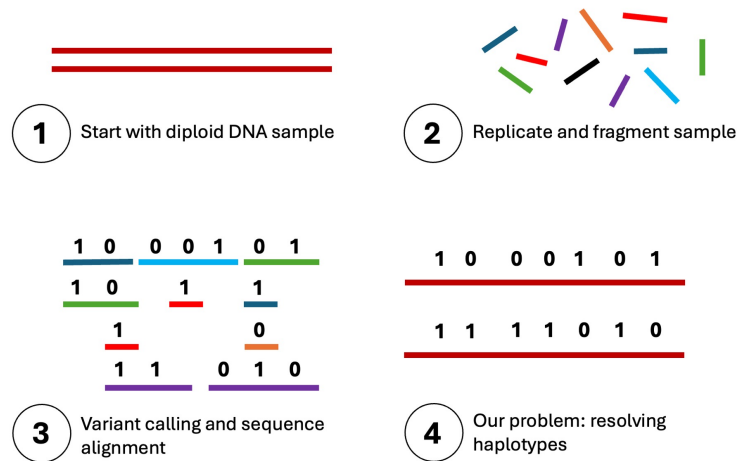


Figure 1: DNA Sequencing Workflow

1.1.2 Importance of Haplotyping

There are a number of applications in which determining this haplotype information is relevant [16]. There is a demonstrated significant phenotypic result of variants which are distributed across two separate haplotypes. Considering haplotype phasing information has helped scientists to learn more about how it affects gene expression, epigenetics, and disease susceptibility.

One instructive example of the importance of haplotyping relates to Alfred Knudson’s two-hit hypothesis [10]. Knudson studied cases of childhood

retinoblastoma, a fast-acting and aggressive cancer, and noticed patterns concerning the hereditary nature of the disease. Through years of studies, he determined that there existed a specific tumor suppressor gene, RB1, related to retinoblastoma. Knudson concluded that a "two-hit" phenomenon was required for development of the tumor. This two-hit hypothesis suggests that both copies of the tumor suppressor gene must possess the mutation which shuts off the suppressive function of the gene in order for the tumor to appear. Individuals who only had the mutation on one copy of the suppressor gene exhibited no phenological difference compared to those without the mutation altogether. It is required for the mutation to be inherited from both parents, or for some epigenetic change to occur to produce the mutation in conjunction with an inherited mutation. Examining haplotypes in order to determine the presence and the source of these types of mutations can help to prevent such diseases and also aids in future studies of genetic disease.

Two distinct techniques are commonly used for resolving haplotypes in a single individual: haplotype assembly and haplotype phasing.

1.1.3 Haplotype Assembly

Most current DNA sequencing techniques first fragment the DNA sample into smaller pieces, then perform polymerase chain reaction (PCR) to repeatedly replicate the DNA fragments. The base pairs on the fragments are read. Then, they align the fragment reads to a reference genome, and perform variant calling to find the SNPs on the genome, and determine which variant the sample expresses. These techniques produce fragments containing SNPs, assumed to be on the same haplotype. The process of Haplotype Assembly pieces together these fragments, reconstructing the two separate haplotypes from a large set of reads [15].

There have been a variety of approaches to the haplotype assembly problem. HapCUT [3], and more recently, HapCUT2 [8] use algorithms which compute max-cuts in graphs constructed using read fragments. HapCompass [1] builds a compass graph containing cycles, and resolves conflicting read evidence to compute haplotypes. HapTree [4] is a method which aims to resolve haplotypes in populations of higher ploidy, or DNA samples with more than two copies of each chromosome. WhatsHap [14] was optimized to consider sequencing error information and increased read length into account.

1.1.4 Haplotype Phasing

Haplotype phasing reconstructs haplotypes using existing haplotype data sampled from a reference population. Phasing requires the allele content to be determined at each SNP, but does not consider these sequence reads as fragments in the same way that haplotype assembly does. Rather, haplotype phasing matches alleles together on the same haplotype based on what is the most probable configuration, determined by the frequency of the haplotypes in the reference population.

Beagle [5] is one of the first haplotype phasing methods developed. Beagle leverages a "localized haplotype-cluster" model, which defines a hidden Markov model, and iteratively samples haplotype pairs to construct complete haplotypes. SHAPEIT2 [7] similarly uses a Markov model for phasing, and achieves competitive performance when compared to Beagle. Eagle2 [13] instead performs phasing using a data structure that was instead based on the Burrows-Wheeler transform. This approach demonstrated considerable speed-up (20x) and accuracy increase (10%) when compared to SHAPEIT2.

1.2 Motivation

Haplotype assembly is a combinatorial method, while haplotype phasing is a probabilistic method. Haplotype assembly is at the mercy of the sequencing read coverage and read quality. A significant shortcoming of haplotype assembly caused by this dependence is that there are often regions of the genome which are unable to be resolved due to insufficient read coverage. Haplotype assembly is accurate, but produces incomplete haplotypes. Haplotype phasing, on the other hand, requires a sufficient reference population of haplotypes, such that there are many haplotypes in the reference panel which are compatible with the target genotype. Additionally, phasing using reference data requires that there is similar reference data. If one were to sequence a new population or a new species, then phasing with a reference panel would be ineffective, due to a lack of similar existing haplotypes.

Haplotype phasing and assembly are two disjoint areas. They are the two leading techniques for single-individual haplotyping, yet they have never been used in conjunction within the same model. Using reference haplotypes can help bridge the gaps left behind during haplotype assembly. Using assembly can help fill reference haplotype databases so that phasing is possible, and more effective in the future.

1.3 Problem Definition

We propose a versatile, undirected graphical model for resolving haplotypes within a single-individual DNA sample. Given a set of sequence reads and genotype-compatible haplotypes sampled from a reference population, our goal is to compute the complete haplotype pair with the highest likelihood.

2 Methods

2.1 Problem Inputs and Preliminaries

The input to the described haplotyping problem is a collection of haplotype-related evidence. Considering current technology, this will be a collection of DNA sequence reads for a single individual, as well as a reference database of compatible haplotypes. These sequence reads are mapped to a reference genome in order to determine the SNP information of the sequences. In haplotype assembly, only heterozygous sites are considered, as homozygous sites are insignificant for those methods. However, for the purpose of the unification method, homozygous sites are included. These can provide important information when considering evidence taken from reference haplotype data. Reads that do not contain sufficient information are excluded. This includes reads that only cover a single SNP site as well as reads which are indecipherable due to sequencing error or incomplete coverage of a SNP site. Reads that cover two or more SNP sites are able to link alleles together on the same haplotype, and therefore are useful for the solution to the haplotyping problem.

We represent SNP information as either a 0 or a 1, where 0 indicates the major allele and 1 indicates the minor allele. We then construct an $m \times n$ matrix, S , such that each row corresponds to a read fragment and each column corresponds to a unique SNP.

2.2 Building the Graphical Model

Our method uses a series of graphical models to represent and solve the haplotyping problem. We initially construct a fragment graph, using the fragment matrix S . A quotient graph is then created from the fragment graph. Then, we attach factors to the nodes and edges of the quotient graph to create a factor graph, which we can then perform inference on.

2.2.1 Fragment Graph

We define the fragment graph for this problem as the undirected graph $G = (V, E)$, such that V is the vertex set where there is one vertex for each unique variant site, or similarly, there is one vertex for each column in the fragment matrix S , and E is the edge set where an edge is constructed between two vertexes if there is at least one fragment which contains both SNPs. This is represented in the fragment matrix if there is a value for the two columns in the same row.

For example, consider an input to the problem with a unique set of read fragments $S = (S_1, S_2), (S_1, S_3), (S_2, S_3, S_4), (S_3, S_4)$. We construct the fragment graph as seen in Figure 2 below.

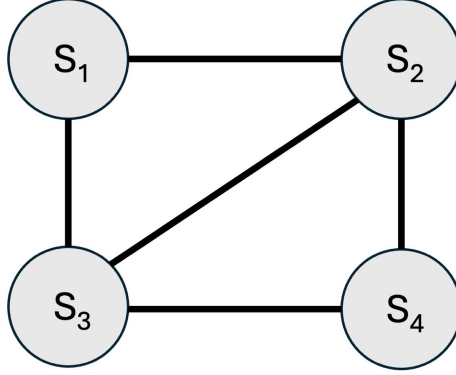


Figure 2: Example Fragment Graph Construction

2.2.2 Quotient Graph

The quotient graph is a partitioning of the fragment graph G . The quotient graph is an undirected graph constructed such that each node in the quotient graph corresponds to two or more different variant sites, and indicates that there is some evidence connecting the phasings at those sites. More formally, each edge between vertices v_i and v_j , $e_{i,j}$, becomes a node, $v_{i,j}$, in the quotient graph. Each clique sized three or greater becomes a single node in the quotient graph representing all SNPs in the clique. The presence of a clique in the fragment graph indicates that there is sufficient evidence between all SNPs included in the clique. For example consider a clique containing v_i , v_j , and v_k . A node is created in the quotient graph representing the three nodes, $v_{i,j,k}$. For every two nodes in the quotient graph which share a variant site, there is an edge constructed between the two. For nodes $v_{i,j}$ and $v_{j,k}$, an edge is created between the nodes, because they share j as a SNP, and can be pieced together in assembly.

Consider again the example where the input to the problem is a unique set of read fragments $S = (S_1, S_2), (S_1, S_3), (S_2, S_3, S_4), (S_3, S_4)$. We construct the quotient graph as shown below in Figure 3.

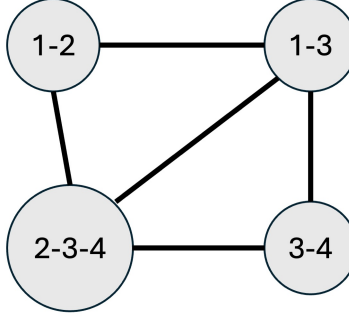


Figure 3: Example Quotient Graph Construction

2.2.3 Factor Graph

We then define factors on the nodes and edges in the quotient graph to produce a factor graph. The factor graph is a model often used for signal processing and propagating "messages" over a graphical model [11]. A factor graph assumes that there is a global function that can be factored into several functions of variables. See Equation 1 for an example factorization for function $f(x, y, z)$:

$$f(x, y, z) = f_1(x, y)f_2(z) \quad (1)$$

We consider the global function of the model to be the probability function for the complete haplotype phasings. We assume this function can be decomposed, or factorized, into local probability functions for each haplotype phasing cluster. These clusters are equivalent to each node in the quotient graph. It follows that we define our factors based on the evidence provided for each haplotype cluster in the quotient graph. We compute all possible phasings for each SNP sequence, which then each become one state in the node factor. We add 1 to a phasing state for each SNP fragment in the DNA sample which indicates the presence of that state. Then, we compute the factors defined on edges as a cross-product of the two node factors which the edge is incident to. Because each edge connects two nodes which share a SNP, we can combine phasing states by joining them on SNPs which are compatible.

2.3 Chordal Contraction

Before we perform inference on the graphical model in order to compute the maximum likelihood phasings, we will first perform operations on the graph to make the graph chordal. A chordal graph is one which does not contain

any simple cycles of size larger than 3. Many inference algorithms performed on graphical models become simpler, more accurate, or faster on graphs which are chordal. We propose a chordal contraction algorithm to achieve this. The primary objective of the algorithm is to reach chordality in the graphical model. We also specify a secondary objective, to preserve entropy, or uncertainty, in the graphical model. The algorithm chooses an edge and contracts it, combining the two nodes which the edge is incident to. This process requires recomputation of the node and nearby edge factors affected by each contraction, as collapsing nodes into longer SNP sequences results in more possible phasings, expanding the node and edge factors exponentially. To solve the exponential growth of node and edge factors, we maintain the 10 greatest potentials in each factor, thus discarding low potential states. This means that we are disregarding haplotype phasings which are already determined to have sufficiently low probability of occurring.

The optimization of chordal contraction is known to be NP-hard, more specifically W2-Hard [2]. An exact algorithm to find the optimal solution, with optimal being the minimum number of edges to contract, is infeasible, as its runtime increases exponentially with the size of the graph. Instead of solving the problem exactly, we propose a heuristic-based algorithm to reach chordality in a number of steps which may not necessarily be optimal. While the graph is not chordal, we select the edge with lowest entropy and contract it, combining the nodes which were the endpoints for the edge.

2.4 Inference

Once the factor graph is made chordal, we begin performing inference on the model in order to compute the probabilities of phasing pairs. The currently proposed algorithm for inference is the junction tree algorithm [9]. This algorithm involves a series of seven steps:

1. Moralization of the graph (unnecessary in our case, as our graphical model is already undirected).
2. Triangulation of the graph (already performed by our chordal contraction algorithm).
3. Form a junction tree.
4. Assign potentials to the junction tree cliques and initialize separator potentials to unity. These potentials are determined by our edge and node factors.
5. Arbitrarily designate one clique to be the root clique.
6. Perform message passing with absorption to and from the root clique until every link on the junction tree has passed a message forwards and backwards.

7. Extract marginal potentials from each clique of the junction tree.

This algorithm presents its own set of challenges and adaptations we must implement in order to efficiently compute the answer. Junction tree requires factors defined upon cliques - these will grow exponential in size, and will require discarding low probability phasings along the way. Once these challenges are resolved, this junction tree and message passing inference algorithm will produce marginal probabilities of phasings, which we can combine to produce the maximum likelihood haplotype phasings.

3 Results

3.1 Proof of Uniqueness

At each iteration of contraction on the graphical model, we collapse nodes with factors distributed across possible phasings into one single node with factors distributed across the combined phasings. To obtain all possible combined phasings, we need an efficient algorithm that considers the phasings identified by the separate nodes. We must also prove that if the phasings of each separate node are unique, then our algorithm produces a set of combined phasings that is also unique. We construct a flow network to demonstrate this property.

We will show that:

1. Each unique set of haplotype phasings corresponds to a unique maximum-flow configuration in flow network $G = (V, E)$.
2. The combination of two unique sets haplotype phasings will result in a unique set of haplotype phasings.

We construct the flow network $G = (V, E)$ for a set of haplotype phasings as such:

- A source node, s , and a sink node, t , are initialized.
- For each SNP in the gene sample, a grid of 4 nodes are created. We label each node p_{ij} such that i indicates whether this is a node in the first or second column, and j indicates the value of the allele (0 or 1 in biallelic gene samples).
- For all j in each grid, an edge is directed from $p_{1,j}$ to $p_{2,j}$. Additionally, an edge is directed from $p_{2,j}$ to all k nodes, p_{1k} , in the next grid.
- An edge is directed from the source to each p_{1j} node in the first grid, and an edge is directed from each p_{ij} node in the final grid to the sink.
- All edges that connect the source and sink to a SNP grid, and connect unique SNP grids together (for example, the colored edges in the example in the figure, edges between SNPs a and b) have infinite capacity.
- Edges within the SNP grids have finite capacity. The capacities are determined by the SNP's genotype. For a heterozygous SNP, p , edges (p_{10}, p_{20}) and (p_{11}, p_{21}) each have unit capacity. For a SNP, p , that is homozygous in allele j , the edge (p_{1j}, p_{2j}) has capacity 2, and the edge $(p_{1\bar{j}}, p_{2\bar{j}})$ has 0 capacity.

We then identify all maximum-flow configurations in the constructed flow network. In this case, sending flow along an edge $a_{2,0}$ to $b_{1,0}$ indicates that one of the haplotype's phasings between the two sites is 0 – 0 (and the other complementary phasing is 1 – 1). See Figure 4 below for an example flow

network representing the possible phasings between two heterozygous sites. We indicate two possible options here for a maximum flow configuration, meaning that there are two possible unique haplotype phasings between these two sites. One is indicated in red and the other in blue.

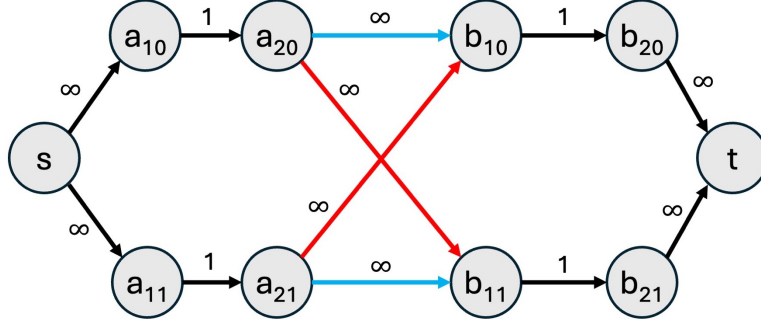


Figure 4: Phasing Flow Network

Between each genetic variant site, represented by the grid of 4 vertices in the flow network, there is a bottleneck of two edges. Each genetic site only connects to the next through two edges - there are no edges that span further than one site. Therefore, altering the maximum flow configuration between two sites cannot change the uniqueness of the maximum flow configurations between any other two sites. This means that joining two sets of unique phasings cannot change the uniqueness of either set, and will produce a combined unique set of phasings.

4 Discussion

The techniques for DNA sequencing and the fields of haplotype assembly and haplotype phasing are constantly changing. NGS emerged in the early 2000s, revolutionizing the industry [18]. This technique produces short-read sequences, which are the data medium of focus in this thesis, as they remain the most common sequencing technology. NGS produces mostly high-quality, accurate reads, and its age has made it a reliable technology. These DNA sequencing reads have made haplotype assembly possible. The building of the International HapMap Consortium [17] allowed the development of haplotype phasing techniques. TGS techniques are becoming more common, and improving in read quality and accuracy [18]. TGS produces much longer read fragments than NGS. The increase in fragment length makes the core problem of haplotype assembly simpler, as there is less piecing together to be done. However, the most significant issues of assembly remain. Errors in variant calling and lack of coverage over certain areas of the genome can result in incomplete haplotype resolution. These shortcomings can be resolved by incorporating reference haplotype information into the solution.

The future of genetics and DNA sequencing is unpredictable. As seen with NGS, TGS, and the HapMap Consortium, methods are constantly emerging, changing, and improving. The model proposed in this thesis is flexible. It can use short-read sequences, long-read sequences, and can cover its gaps with reference haplotype information. For any future developments in DNA sequencing, this model will be able to incorporate the new data input, so long as it can be adapted into some "fragment"-like structure, similarly to how we adapt reference haplotypes to be full-sized "fragments". Our novel approach to the problem, unifying haplotype phasing and haplotype assembly, achieves versatility, completeness, and longevity in the field.

There is significant future work to be performed using this model. We plan on achieving results on real-world data, ensuring that our model is competitive with the top current methods for haplotype assembly and haplotype phasing. Because this is a novel method, there is likely significant efforts that can be taken towards the optimization of its accuracy and runtime. It may not outperform other models to begin with, but if it is competitive in accuracy and speed, its versatility will prove its value. This versatility can also be extended to approach the polyploid haplotype phasing problem. Our current model only supports diploid samples, but it can be easily adapted to accomodate polyploid phasing. This is an area where few current methods exist, and our method can address a significant knowledge gap in the area.

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